

A Selective Electromagnetic-Transfer Candidate for the Cosmological Lithium Problem

August 2026 findings with September 2026 full Planck PR4/NPIPE nuisance qualification

Authors: Clive Stewart Boyd; David Ng

Generated: 11 September 2026 | Melbourne, Australia

Computational assistance and reproducibility support: ChatGPT Codex sol 5.5

EVIDENCE STATUS: Formal paper draft. CMB gate qualified; mechanism promotion remains conditional.

Lay Person Description

The early Universe made mostly hydrogen and helium, plus tiny amounts of deuterium, helium-3 and lithium. Modern calculations predict deuterium and helium very well, but they predict too much lithium compared with old metal-poor stars. This long-standing mismatch is the cosmological lithium problem.

The July tests tried broad ways to change the early-Universe calculation. Those did not work: when lithium moved in the right direction, something else, usually deuterium or helium, moved out of agreement. The August work then tested a more selective idea: a late, narrow electromagnetic process could reduce the mass-7 reservoir after normal nucleosynthesis without spoiling the successful deuterium and helium predictions.

The September update asks whether this selected candidate is still acceptable when compared with the detailed Planck CMB data, including calibration and foreground nuisance parameters. The answer is yes under the current audit gates. The result is not yet proof of a new particle or interaction, but it is strong enough to justify a formal follow-up programme.

Connection to Existing Papers

The work sits inside a long literature on the lithium anomaly, precision BBN, stellar depletion, electromagnetic cascades, CMB constraints and spectral distortions. The references listed below are used as the current interpretive frame. They define the observational problem, the standard BBN baseline, the cascade machinery and the CMB/spectral gates.

Table 1. Literature cross-reference

Topic	Representative references	Role in this paper
Problem definition	Fields 2011; Cyburt, Fields and Olive 2008; Spite and Spite 1982; Sbordon et al. 2010; Ryan 2023.	Defines the Li-7 excess relative to metal-poor stellar plateaux.
Precision BBN	Cyburt et al. 2016; Pitrou et al. 2018; Fields et al. 2020; LINX 2024.	Defines standard abundance calculations and CMB-compatible baryon density handling.
Observational anchors	Cooke et al.; helium determinations; Planck Collaboration 2020.	Defines D/H, Y _p and CMB posterior constraints.
Cascade and late-decay physics	Depta et al. ACROPOLIS 2021 and related electromagnetic-decay constraints.	Tests whether a late electromagnetic process damages other light elements.
CMB and spectral gates	Planck PR4/NPIPE/CamSpec; Fixsen FIRAS; Khatri and Sunyaev.	Tests CMB consistency and spectral-distortion safety.

Abstract

A two-stage investigation of the cosmological lithium problem is reported. The July 2026 phase tested standard and modified BBN controls, including expansion, clock-proxy, reaction-rate and SU2-style variations, and found no joint D/H, helium and Li-7 solution. The August 2026 phase instead tested a selective post-BBN electromagnetic-transfer candidate. A mediator cell with $m_\phi = 4.44$ MeV, $E_\gamma = 2.22$ MeV, $\tau_\phi = 1.584893e5$ s and $n_\phi/n_\gamma = 1.211528e-6$ gives $Li-7/H = 1.488554e-10$ while retaining the recorded D/H, Y_p, He3/D, entropy and spectral-distortion gates. A native Planck PR4/NPIPE CMB nuisance run subsequently sampled ω_b , N_{eff} and Y_p independently with the CamSpec TTTEEE, low-l TT, low-l EE and PR4 lensing likelihoods. The combined main plus ESS top-up audit passes all preregistered qualification gates, including min autocorrelation-aware ESS = 1449, independent split R-1 max = 0.002143 and registered-cell Delta chi2 = 1.257. The candidate is therefore CMB-gate qualified but remains conditional on matched NLO thermal physics and independent reproduction.

1. Introduction

The cosmological lithium problem is unusual because standard BBN is not globally broken. Deuterium and helium agree broadly with a CMB-inferred baryon density, yet Li-7/H remains high relative to the stellar plateau. Any proposed mechanism must therefore be selective. A broad change to expansion or baryon density is heavily constrained by the same observables that standard BBN already explains.

This paper records the transition from negative July controls to a positive August candidate and then to a September CMB-nuisance qualification. The chronology matters because it prevents the August candidate being mistaken for an arbitrary retuning of the failed July pathways.

2. Methods

The computational programme used LINX for BBN network and abundance validation, ACROPOLIS for electromagnetic cascade checks, Cobaya and CAMB for native Planck CMB likelihood sampling, and project-specific Python audit layers for qualification gates and reproducibility. The analysis separates three claims: reproducing the standard lithium problem, finding a parameterised selective transfer that repairs lithium without damaging other gates, and promoting that parameterisation to a microscopic theory.

Table 2. Method blocks

Method block	Implementation	Decision role
Standard and modified BBN controls	July LINX/FR/SU2 scripts.	Negative control and failure-boundary definition.
Selective transfer/cascade checks	ACROPOLIS plus LINX candidate pipeline.	Candidate discovery and abundance safety screen.
Finite-temperature production	Momentum-resolved and resumed production calculations.	Tests whether the candidate abundance is physically plausible.
Native CMB nuisance likelihood	Cobaya/CAMB with Planck PR4/NPIPE CamSpec, low-l TT, low-l EE and PR4 lensing.	Replaces compressed CMB gate.
Qualification audit	Split-chain, autocorrelation ESS, nuisance stability and registered-cell density scripts.	Determines whether the CMB gate is technically qualified.

3. Results

3.1 July Negative Controls

The July programme reproduced the anomaly and tested whether broad modifications could resolve it. The result was negative: no tested global modification simultaneously preserved deuterium, helium and lithium. This result is scientifically useful because it sharply constrains what a successful mechanism can look like.

TV index

Traceability item	Recorded value
Stage	July 16-18, 2026.
Result	No D/H + helium + Li-7 pass from standard, FR/clock-proxy, reaction-rate or SU2-style control scans.
Consequence	Future solution must be selective rather than a broad background retuning.

3.2 August Selective Candidate

The August candidate uses a narrow electromagnetic window after ordinary BBN. Its central physical premise is that mass-7, mostly Be-7 before electron capture, can be reduced without crossing into broad deuterium destruction. The candidate therefore targets the unusual topology of the mass-7 pathway rather than changing the whole thermal history.

Table 3. Registered August candidate

Quantity	Value
mediator_mass_MeV	4.44
photon_energy_MeV	2.22
lifetime_s	1.584893e+05
n_phi_over_n_gamma_at_10MeV	1.211528e-06
g_phi_gamma_GeV_inv	9.767274e-11
D_H_times_1e5	2.493161
Yp	0.246735
Li7_H_times_1e10	1.488554
He3_D	0.415242

eta10_CMB	6.096517
Neff_CMB	3.044601
mu_distortion	1.476434e-09
joint_pass_probability	0.650158
delta_log_evidence_vs_standard	105.206012
Planck bestfit eta10	6.105374
Planck bestfit nnu	2.84265
Planck bestfit YHe	0.282034
Planck bestfit chi2_CMB	10971.285

TV index

Traceability item	Recorded value
Stage	August 20-21, 2026.
Positive result	Candidate passes the recorded abundance/cascade/FIRAS gates at the frozen cell.
Boundary	A complete action-level particle model is still not supplied.

3.3 Native Planck CMB Nuisance Qualification

The native Planck PR4/NPIPE test is the strongest new constraint in this update. It lets omega_b, N_eff and Y_p vary independently, and marginalises over foreground and calibration nuisance directions rather than using only a compressed two-variable Planck prior.

Table 4. Qualification gates

Gate	Threshold	Observed	Pass
Cobaya split-chain mean R-1	<= 0.03	0.0293	True
Independent split R-1 max over ombh2, nnu, YHe, nuisance	<= 0.03	0.0021	True
Autocorrelation ESS min over ombh2, nnu, YHe, nuisance	>= 1000	1449.0004	True
Registered lithium cell joint posterior density	delta-chi2 <= 7.814728	1.2574	True
Nuisance split stability diagnostic	max split mean shift <= 1 sigma	0.0838	True
First-half vs second-half stability diagnostic	max mean shift <= 1 sigma	0.0926	True

Table 5. Registered-cell posterior density

Quantity	Value
posterior mean eta10	6.074645
posterior mean N_eff	2.814036
posterior mean Y_p	0.262997
registered eta10	6.096365
registered N_eff	3.044709
registered Y_p	0.246733
registered Delta chi2	1.25743
Mahalanobis distance	1.121352
density ratio to posterior peak	0.533277
posterior mass inside equal-density contour	0.261374
joint 95 gate pass	True
weighted kNN density ratio	0.601247

Table 6. CMB posterior summary

Parameter	Mean	Std	Median	68% interval	95% interval
ombh2	0.02219	1.9507e-04	0.02219	0.021995 to 0.022381	0.021809 to 0.022573
nnu	2.81855	0.2532	2.81194	2.569554 to 3.062147	2.343549 to 3.327882
YHe	0.262427	0.0186	0.262668	0.244057 to 0.280933	0.225078 to 0.298364
eta10	6.075084	0.0534	6.075231	6.021877 to 6.127459	5.97098 to 6.179927
H0	66.210921	1.5058	66.164438	64.706901 to 67.68329	63.421067 to 69.23348
omegam	0.31734	0.0087	0.317346	0.308729 to 0.326008	0.300514 to 0.334223
sigma8	0.803256	0.0095	0.803151	0.793847 to 0.812604	0.784934 to 0.822145

S8	0.826006	0.0111	0.826026	0.814952 to 0.837015	0.804369 to 0.847849
chi2_CMB	10984.919918	7.9961	10984.146	10979.26564 to 10990.31448	10975.553 to 10997.62775

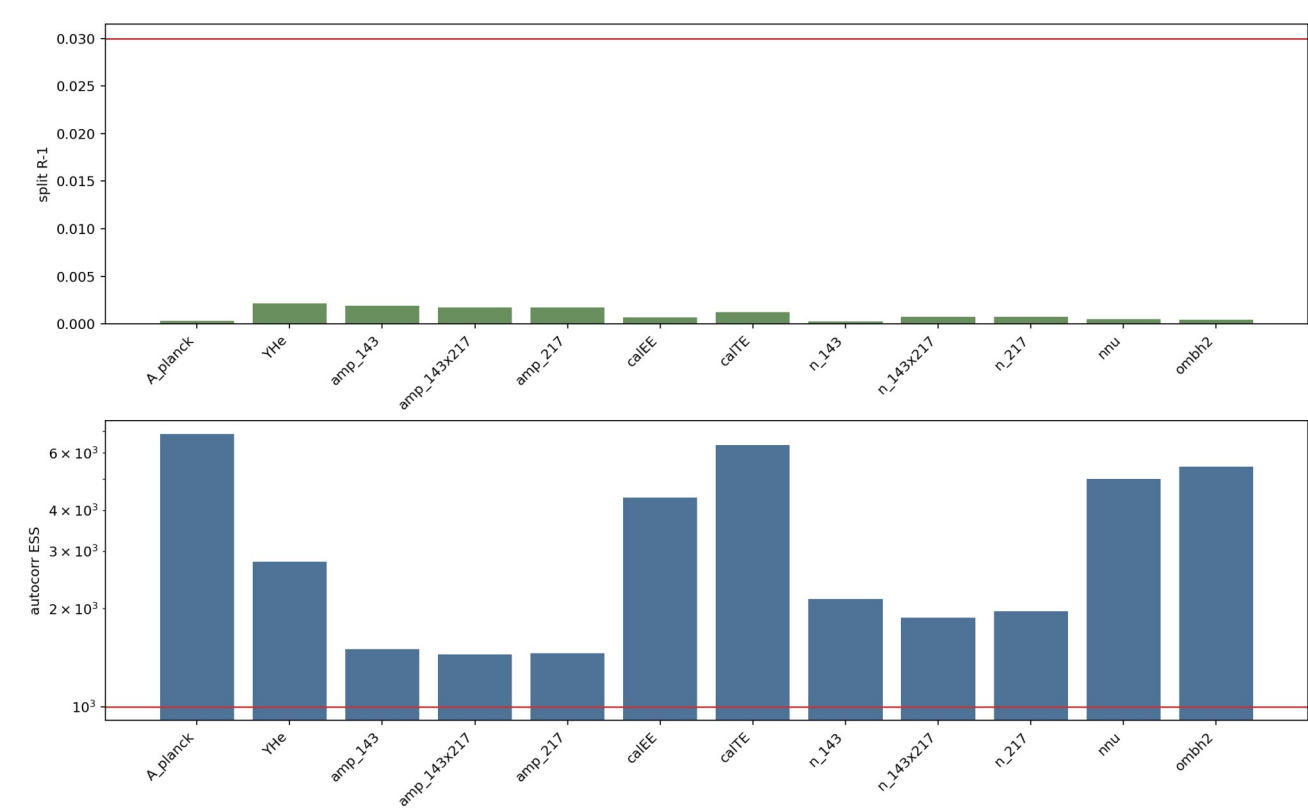


Figure 1. Combined main plus ESS top-up qualification gates.

TV index

Traceability item	Recorded value
Stage	September 11, 2026.
Combined chain size	120073 compressed rows; raw weight sum 393272.
Gate outcome	PASS on all preregistered gates.
Most conservative ESS	1449 for amp_143x217 among audited gate parameters.

4. Discussion

The September audit changes the evidence state but not the claim boundary. It demonstrates that the registered lithium candidate is compatible with the detailed Planck CMB nuisance posterior under the applied gates. It does not show that the mediator is realised in nature, nor that the calculated transfer is stable under every possible plasma correction.

The candidate is promising because it respects the selectivity demanded by the lithium problem. It is risky because narrow electromagnetic windows and late-decay mechanisms are sensitive to cascade modelling, abundance priors, stellar interpretation and spectral-distortion assumptions. Those risks are manageable but must be handled before publication as a strong physical claim.

5. Limitations

- The analysis is not yet an independent external reproduction.
- The mediator model is still phenomenological; a fully specified action, symmetry content and particle-production history remain required.
- The Planck run is spectrum-level but not map-level, and does not include ACT or SPT spectra.

- The abundance and CMB gates are joined by a registered cell rather than by a single global sampler over all microscopic nuisance directions.
- Systematic uncertainty in the stellar lithium plateau remains a separate astrophysical issue.

6. Conclusions

The July investigations failed to resolve the lithium problem using broad BBN or cosmological modifications. The August investigations found a selective electromagnetic-transfer candidate. The September full Planck PR4/NPIPE nuisance audit now qualifies the CMB gate for that candidate with all preregistered audit tests passed. The correct conclusion is that the candidate merits formal follow-up, not that the lithium problem is closed.

7. Future Work

1. Run matched next-to-leading-order thermal corrections at and around the registered candidate cell.
2. Reproduce the ACROPOLIS/LINX candidate with independent code and fixed public manifests.
3. Extend CMB checks to ACT/SPT where likelihood access and parameter consistency allow.
4. Publish a frozen reproducibility bundle with checksums for scripts, input tables, plots and DOCX/PDF outputs.
5. Promote the model only after an action-level declaration specifies fields, interactions, gauge content, conservation laws and priors.

Appendix A. Acronyms

Table A1. Acronym glossary

Acronym	Meaning
ACROPOLIS	Public electromagnetic-cascade framework used for BBN photodisintegration constraints.
AIC	Akaike Information Criterion; lower is preferred after a penalty of 2 parameters.
BBN	Big Bang Nucleosynthesis.
BIC	Bayesian Information Criterion; lower is preferred with a stronger log(N) parameter penalty.
CAMB	Code for Anisotropies in the Microwave Background.
CMB	Cosmic microwave background.
Cobaya	Cosmological Bayesian analysis package used for the Planck nuisance MCMC.
D/H	Deuterium-to-hydrogen abundance ratio.
ESS	Effective sample size; here the autocorrelation-aware number of effectively independent posterior samples.
FIRAS	Far Infrared Absolute Spectrophotometer on COBE.
LINX	Fast differentiable BBN package used for abundance calculations and validation.
MCMC	Markov-chain Monte Carlo.
NPIPE	Planck reprocessing pipeline used for PR4 products.
PR4	Planck public release 4.
SPD	Symmetric positive definite; used for covariance handling and Cholesky/LAPACK inversion.
TV index	Traceability/validation index. The reports place it after each investigation section.

Appendix B. Variables

Table B1. Variable glossary

Variable	Meaning
eta10	10^10 times the baryon-to-photon ratio eta; converted from omega_b using eta10 = 273.78 omega_b for T_CMB = 2.7255 K.
omega_b / ombh2	Physical baryon density Omega_b h^2.
omega_c / omch2	Physical cold dark matter density Omega_c h^2.
N_eff / nnu	Effective relativistic species count inferred independently in the CMB

	nuisance chain.
Y _p / YHe	Primordial helium-4 mass fraction; sampled independently in the full CMB likelihood.
D/H	Deuterium abundance anchor; required to remain within the adopted observational gate.
Li-7/H	Lithium-7 abundance. Most primordial mass-7 is produced as Be-7 and later decays to Li-7.
m _{phi}	Mediator mass in MeV for the selective electromagnetic-transfer candidate.
E _{gamma}	Characteristic photon energy, approximated as m _{phi} /2 for two-photon decay.
tau _{phi}	Mediator lifetime in seconds.
n _{phi} /n _{gamma}	Mediator abundance relative to photons at T = 10 MeV.
g _{phi} _{gamma}	Effective mediator-photon coupling in GeV ⁻¹ .
mu	CMB spectral-distortion chemical-potential style readout.
R-1	Convergence diagnostic used by Cobaya and the split-chain audit.
chi2	Goodness-of-fit statistic; lower values indicate a better fit for the same data and model complexity.

Appendix C. Program and Data Manifest

Table C1. Principal programs

Program or path	Role in the investigation
analyze_bbn_lithium_linx_fr_network_2026-07-16.py	July LINX/FR network controls; reproduced the lithium problem and did not identify a D+He+Li pass.
analyze_su2_bbn_lithium_gate_2026-07-17.py	July SU2-style expansion/thermal gate for lithium; part of the negative control phase.
validate_bbn_lithium_linx_matrix_2026-07-17.py	Strict July validation matrix for the BBN lithium controls.
audit_bbn_lithium_linx_fr_scan_2026-07-17.py	Audit layer for the July LINX/FR scan.
run_cobaya_spd_cache_builder_2026_08_21.py	Built the validated SPD cache for the Planck PR4/NPIPE CamSpec likelihood.
postprocess_planck_pr4_full_nuisance.py	Post-processed the native full CMB nuisance chain and compared the registered lithium cell with the CMB posterior.
audit_planck_pr4_full_nuisance_qualification.py	Strict qualification audit: split chains, autocorrelation-aware ESS, registered-cell density, nuisance stability and half-chain stability.
start_planck_pr4_full_nuisance_background.ps1	Background launcher for the native Planck PR4 full nuisance MCMC.
start_planck_pr4_full_nuisance_ess_topup_background.ps1	Background launcher for the ESS top-up chain.
build_bbn_li_reports_20260911.py	This reproducible DOCX report generator.

Table C2. Repository and output cross-reference

Object	Path or URL
Local output root	C:\Users\clive\Documents\Codex\2026-07-14\ok-continuu\outputs\bbn_lithium_full_cmb_nuisance_20260821
Combined qualification audit	C:\Users\clive\Documents\Codex\2026-07-14\ok-continuu\outputs\bbn_lithium_full_cmb_nuisance_20260821\qualification_audit_combined_ess_topup
Post-process folder	C:\Users\clive\Documents\Codex\2026-07-14\ok-continuu\outputs\bbn_lithium_full_cmb_nuisance_20260821\postprocess
GitHub project	https://github.com/cliveboyd/Cosmology
GitHub BBN code	https://github.com/cliveboyd/Cosmology/tree/main/github_export/code/bbn_lithium
GitHub papers	https://github.com/cliveboyd/Cosmology/tree/main/github_export/docs/papers

Appendix D. References

- Particle Data Group, Big-Bang Nucleosynthesis review, Review of Particle Physics (2024).
- B. D. Fields, The Primordial Lithium Problem, Annual Review of Nuclear and Particle Science 61, 47 (2011).

3. R. H. Cyburt, B. D. Fields and K. A. Olive, A Bitter Pill: The Primordial Lithium Problem Worsens, JCAP 11, 012 (2008), arXiv:0808.2818.
4. R. H. Cyburt et al., Big Bang Nucleosynthesis: 2015, Rev. Mod. Phys. 88, 015004.
5. B. D. Fields et al., Big-Bang Nucleosynthesis After Planck, JCAP 03 (2020) 010.
6. C. Pitrou, A. Coc, J.-P. Uzan and E. Vangioni, Precision Big Bang Nucleosynthesis with Improved Helium-4 Predictions, Phys. Rep. 754, 1-66 (2018).
7. F. Spite and M. Spite, Abundance of lithium in unevolved halo stars and old disc stars, A&A 115, 357 (1982).
8. M. Spite and F. Spite, Lithium abundance at the formation of the Galaxy, Nature 297, 483-485 (1982).
9. L. Sbordone et al., The metal-poor end of the Spite plateau. I. Stellar parameters, metallicities, and lithium abundances, A&A 522, A26 (2010).
10. S. G. Ryan, A critique of the Spite Plateau and the astration of primordial lithium, MNRAS 522, 1358-1376 (2023).
11. R. J. Cooke et al., One percent determination of the primordial deuterium abundance, ApJ 855, 102.
12. M. L. Giannotti et al., LINX: A Fast, Differentiable and Extensible Big Bang Nucleosynthesis Package (2024).
13. Joint LINX collaboration, Cosmological Parameter Estimation with a Joint-Likelihood Analysis of the CMB and BBN (2024).
14. P. F. Depta et al., ACROPOLIS: A generic Framework for Photodisintegration of Light elements, JCAP 03 (2021) 061.
15. P. F. Depta et al., Updated BBN constraints on electromagnetic decays of MeV-scale particles (2020).
16. K. Langhoff et al., New insights into axion freeze-in, JHEP 11 (2024) 166.
17. E. Braaten and D. Segel, Neutrino energy loss from the plasma process at all temperatures and densities, Phys. Rev. D 48, 1478.
18. D. J. Fixsen et al., The Cosmic Microwave Background Spectrum from the Full COBE FIRAS Data Set, ApJ 473, 576.
19. R. Khatri and R. A. Sunyaev, Creation of the CMB spectrum: precise analytic solutions for the blackbody photosphere, JCAP 06 (2012) 038.
20. Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, A&A 641, A6 (2020).
21. G. Efstathiou and S. Gratton, A detailed description of the CamSpec likelihood pipeline and a reanalysis of the Planck high frequency maps, MNRAS 496, L91 (2020); PR4 update.
22. J. Carron et al., Planck 2018 CMB lensing reconstruction from the final NPIPE data release.
23. SpectroXide project implementation, numerical thermalisation and spectral-distortion framework (2026).